Aniruddha Mukherjee

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EDUCATION

Master of Technology, Biomedical Engineering Autumn 2025–Present

Biosciences and Bioengineering, Indian Institute of Technology Bombay, India
Advisors: Professor Saket Choudhary and Professor Shamik Sen

Bachelor of Technology, Computer Science and Engineering Autumn 2021–Spring 2025

School of Engineering, Tezpur University, India
Mentor: Professor Nityananda Sarma

RESEARCH PROJECTS

Seqout December 2025–Present

Search engine for public genomic data spanning seven major repositories, featuring LLM-based sample metadata harmonisation, FASTQ read-derived sample characterisation and microbial content detection using unitigs, and an MCP server for AI agents. Available at seqout.org.

Scintilla September 2025–Present

Hardware-accelerated negative binomial regression-based normalisation for single-cell RNA-seq counts, with support for NVIDIA CUDA, Apple MLX, and multithreaded CPU execution via Numba. Scanpy and Seurat compatible Python and R packages. Work currently private.

Lipika 2024

Small GPT-2 architecture-based language model pre-trained in Assamese (a low-resource language), with custom Rust-based pipeline (github.com/amkhrjee/wp) for scraping data off Wikipedia at scale. Developed as part of bachelor's thesis at Tezpur University.

PERSONAL PROJECTS

TinyPy 2024

Zero-dependency Python package for linear algebra, compatible with NumPy, supporting rank computation, eigenvalue calculation, and linear system solving. Available at github.com/amkhrjee/tinypy.

Pathfinder 2024

Desktop GUI application written in Go to visualise classic AI algorithms, including A* search, local beam search, and Dijkstra's algorithm. Available at github.com/amkhrjee/pathfinder.

Lox 2024

Implementation of Lox, a dynamically typed programming language. Compiler written in C. Interpreter written in Java. Available at https://github.com/amkhrjee/loxlang.

Pygraphia 2023

Zero-dependency Python package for graph data structures and algorithms, supporting breadth-first search, depth-first search, and minimum spanning tree generation. Available on PyPI at pypi.org/project/pygraphia.

OCWD 2023

Bash-based command-line tool for bulk downloading course material from MIT OpenCourseWare. Over 1,000 downloads and 90 GitHub stars. Available at github.com/amkhrjee/ocwd.

Fontonic 2023

TypeScript-based browser extension for customising website typography, designed to improve reading accessibility. Over 10,000 daily users and over 50,000 installs. Available at github.com/amkhrjee/fontonic.

EXPERIENCE

Teaching Assistant August 2026–Present

Autumn 2026: Programming for Biomedical Data Analysis (BB 709)

I teach MATLAB through mathematical modelling to master's and PhD students with Prof. Deepak Agrawal. Code used in sessions are available at github.com/amkhrjee/BB709.

Research Intern December 2024–January 2025

Department of Computer Science and Engineering, Indian Institute of Technology Patna
Mentor: Professor Rajiv Misra

Implemented reinforcement learning models in PyTorch for resource allocation in mobile edge computing, to support benchmarking for an ongoing research study at the Metaverse lab.

Full-stack Engineering Intern September 2020–November 2020

Foodable.in, New Delhi, India (Remote)

Developed frontend, backend, and database components for this food-ordering startup, collaborating with a three-member engineering team.

PUBLICATIONS

Aniruddha Mukherjee, Mukesh Reddy and Saket Choudhary "Seqout: metadata harmonisation for public genomic data discovery" (in submission; preprint available at seqout.org/paper)

TALKS AND POSTER PRESENTATIONS

1. Aniruddha Mukherjee, Mukesh Reddy and Saket Choudhary. "Seqout: search engine for the world's genomic data". Talk presented at the 6th IndiaFOSS conference, September 2026, Bengaluru, India. doi.org/10.5281/zenodo.23242105
2. Aniruddha Mukherjee, Mukesh Reddy and Saket Choudhary. "Seqout: metadata harmonisation for public genomic data discovery". Poster presentations at:
   - The 6th Genomics India Conference (GIC), September 2026, Bengaluru, India doi.org/10.5281/zenodo.23239647
   - Mathematics of the real world: Research–Industry–Academia Workshop, August 2026, IIT Bombay, India. doi.org/10.5281/zenodo.23241736
   - CSHL Biological Data Science Conference, November 2026, Cold Spring Harbor, New York, USA (accepted)
   - ACM IKDD 14th International Conference on Data Science (CODS), December 2026, Gandhinagar, India (accepted)

RESEARCH MENTORING

1. Mukesh Reddy, Bachelor of Technology, Biotechnology, National Institute of Technology Warangal. Mentored during a summer research internship, May 11–July 15, 2026. Contributed to streamlining Seqout's data ingestion pipeline and optimising the rendering performance of its 2D UMAP dataset embedding visualisation.
2. Harshith Pandela, Bachelor of Technology, Electrical Engineering, Indian Institute of Technology Bombay. Co-mentored an R&D project with Prof. Saket Choudhary, Spring 2025, exploring GPU acceleration for generalised linear models implemented in R.

CERTIFICATIONS

- AWS Certified Cloud Practitioner. Issued by Amazon Web Services, August 2024.